
CORRECTION FORMAT AND THE CONTINUED INFLUENCE OF MISINFORMATION: BELIEF CHANGE AS A MEDIATING MECHANISM

[ANONYMIZED FOR REVIEW]

ABSTRACT

Misinformation often continues to influence reasoning even after corrections are provided, a phenomenon known as the continued influence effect (CIE). The format in which corrections are delivered may affect their efficacy, but the mechanisms through which correction format reduces CIE remain incompletely understood. We tested whether the format of corrective information—standard fact-check, truth-sandwich, or no correction—influences misinformation-consistent inferences and whether belief change mediates this relationship. Participants ($N = 529$) recruited via Prolific Academic read 12 health and science articles with embedded misinformation, each paired with a correction in one of three formats (within-subjects), and completed an inference test either immediately or after a self-scheduled delay (between-subjects). Both active correction formats significantly reduced misinformation-consistent inferences relative to no correction in within-subjects analyses (fact-check: $d = -0.49$, $p < .001$; truth-sandwich: $d = -0.38$, $p < .001$), though the between-subjects main effect of correction format was not significant, $F(2, 523) = 1.49$, $p = .228$, $\eta_p^2 = .006$. Belief change significantly mediated the effect of correction format on inferences; both indirect effects' 95% bias-corrected bootstrap CIs excluded zero. A significant delay main effect emerged, $F(1, 523) = 21.81$, $p < .001$, $\eta_p^2 = .040$, with lower CIE at delayed test. These findings indicate that correction format influences CIE through belief updating, but detection of these effects depends critically on design sensitivity.

Keywords continued influence effect; misinformation correction; belief change; fact-check; truth sandwich

1 Introduction

The persistence of misinformation in public discourse poses a significant challenge to informed decision-making in domains ranging from public health to political participation. Even when false claims are explicitly corrected, they often continue to shape people's reasoning, a robustly replicated phenomenon known as the continued influence effect (CIE; ??). Understanding what makes corrections effective is therefore a priority not only for cognitive theory but also for practitioners designing interventions to counter misinformation. The present study addresses a gap in this literature: whether the format in which corrections are delivered influences their ability to reduce CIE, and through what mechanism.

The CIE has been replicated across more than 50 studies using varied stimuli and populations (?). Corrections reliably reduce belief in misinformation relative to no-correction controls, yet they rarely eliminate its influence on inferential reasoning. For example, ? found that explicit warnings before a correction reduced but did not eliminate CIE, and ?, in a meta-analysis of debunking interventions, concluded that corrections produce moderate effects on belief ($d = 0.35$ – 0.50) but leave residual CIE. The format of the correction matters: the “truth sandwich” approach—which frames a correction by first warning about misinformation tactics, then stating the fact, then providing a positive takeaway—has been proposed as more effective than a standard fact-check (?), though direct comparative evidence remains limited.

What remains unresolved is the mechanism through which correction format affects CIE. Dual-process models of persuasion (?) suggest that corrections that engage more elaborate processing should produce stronger and more persistent belief change. The source monitoring framework (?) further predicts that the ability to attribute a claim

to its correct source (i.e., the correction rather than the misinformation) is critical for reducing CIE. However, few studies have directly measured whether belief change mediates the effect of correction format on inferential reasoning. Additionally, the role of test delay is unclear: some evidence suggests CIE persists or even strengthens over time (consistent with the sleeper effect; ?), whereas other work finds that CIE diminishes with delay as memory for both the misinformation and its correction fades (?).

The current study addresses two open questions. First, do active correction formats (fact-check and truth-sandwich) reduce misinformation-consistent inferences more than a no-correction control? Second, is this effect mediated by belief change? We conducted a secondary analysis of a publicly available dataset from ? (Experiment 2) that manipulated correction format (within-subjects) and test delay (between-subjects) in a large sample ($N = 529$). The data allow us to test the following pre-registered hypotheses:

H1: Correction format will affect misinformation-consistent inferences, with both fact-check and truth-sandwich formats producing lower inference scores than the no-correction control condition.

H2: Myth belief change (pre-correction to post-correction) will mediate the effect of correction format on misinformation-consistent inferences. The indirect path is: correction format $\rightarrow$ greater belief change $\rightarrow$ fewer misinformation-consistent inferences.

H3 (exploratory): Test delay will moderate the effect of correction format on inference scores, such that the advantage of active correction formats over control may differ between immediate and delayed testing.

2 Method

2.1 Participants

We analyzed data from ? (Experiment 2), retrieved from the Open Science Framework (OSF project shf63). The original study recruited 529 adults through Prolific Academic (284 assigned to the immediate test condition, 245 to the delayed condition). After applying exclusion criteria (Session 1 duration < 5 min or > 3 hr, or $|z| > 3$ on inference or pre-belief scores), 13 participants were removed, yielding a final analytic sample of 516 participants ($M_{\text{age}} = 46.39$, $SD = 15.34$, range = 19–85). The sample included 269 women, 242 men, 4 participants who self-described their gender, and 1 who preferred not to disclose. Educationally, 183 participants held a 4-year college degree, 121 had some college without a degree, 86 held a postgraduate degree, 60 were high school graduates, 58 held a 2-year associate degree, and 8 had not completed high school. Participants received monetary compensation (approximately £10.00/hr equivalent). The sample was predominantly WEIRD (Western, Educated, Industrialized, Rich, Democratic); drawn from the U.S. and U.K. Prolific pools, these findings may not generalize to non-WEIRD populations.

2.2 Materials

Stimuli consisted of 12 scenarios adapted from real-world health and science misinformation claims (e.g., vaccines, fluoridation, natural remedy efficacy). Each scenario included a brief news article (~ 150 words) containing an embedded false claim and a subsequent corrective statement presented in one of three formats manipulated within-subjects. The three correction formats were: (a) control—no correction provided after the article; (b) fact-check only—a standard factual correction stating that the claim was false and providing accurate information; and (c) truth sandwich—a correction preceded by a warning about misinformation tactics and followed by a positive key takeaway, as defined by ?. Each participant read all 12 scenarios (4 per format condition), with format assignment counterbalanced across scenarios.

The primary outcome was a bespoke inference test assessing misinformation-consistent reasoning ($\alpha = .79$, as reported by Swire-Thompson et al., 2024). Participants answered 12 scenario-specific questions on an 11-point scale (0 = not at all likely to 10 = extremely likely); higher scores indicate greater reliance on misinformation despite corrections. Belief ratings for each myth and fact claim were collected on the same 0–10 scale both before and after the correction phase, allowing computation of belief change scores. The Comprehensive Assessment of MisBeliefs Index (CAMBI; ?) was administered as a measure of general misinformation susceptibility (17 items, 1–7 scale, $\alpha = .82$). All measures are described in detail in the original publication (?).

2.3 Procedure

Participants completed the study online via Prolific Academic. In Session 1, after providing informed consent and demographic information, participants rated their baseline belief in each of 12 myth claims and 12 factual claims. They then read 12 news articles (one per screen, presented in randomized order), each containing a correction in one

of the three formats. Immediately after reading, participants again rated their belief in each claim (post-correction belief). The inference test—the primary measure of CIE—was then administered, followed by the CAMBI scale. For participants in the immediate condition, the session concluded with debriefing. For participants in the delayed condition, Session 2 was self-scheduled within a 7-day window and consisted of a re-administration of the belief ratings and inference test followed by debriefing. Median Session 1 duration was 816 s (~13.6 min); median Session 2 duration was 381 s (~6.4 min). No attention checks or manipulation checks were included in the original protocol; duration-based engagement screening was applied as a proxy.

2.4 Design

The experiment used a 3 (correction format: control vs. fact-check only vs. truth sandwich) $\times$ 2 (test delay: immediate vs. delayed) mixed factorial design. Correction format was manipulated within-subjects (each participant saw all three formats across counterbalanced scenarios). Test delay was manipulated between-subjects (participants were assigned to either immediate or delayed testing). The primary dependent variable was the misinformation-consistent inference score (0–10 scale). Myth belief change (post-correction belief minus pre-correction belief) was treated as a mediator. To align the analysis with a between-subjects analytical framework (consistent with the pre-registered design in the original research plan), our primary analysis extracted each participant’s matched outcome from the single correction-format condition to which each scenario set was assigned, treating correction format as a between-subjects factor for inferential testing. The within-subjects structure was preserved in supplementary robustness analyses.

3 Results

3.1 Descriptive Statistics and Manipulation Checks

The full sample ($N = 529$) showed a mean misinformation-consistent inference score of 5.74 ($SD = 2.12$) on the 0–10 scale, indicating moderate CIE overall (at approximately the scale midpoint). Mean pre-correction myth belief was 5.53 ($SD = 1.76$), and corrections produced a mean belief reduction of -2.05 ($SD = 2.60$) across all conditions. CAMBI scores ($M = 5.22$, $SD = 0.69$) indicated moderate-to-high general susceptibility to misinformation beliefs, comparable to previous norms. Randomization checks confirmed that participants did not differ across the three correction-format conditions on age, $F(2, 526) = 0.02$, $p = .976$; gender, $\chi^2(6) = 3.16$, $p = .789$; education, $\chi^2(10) = 11.25$, $p = .338$; or political affiliation, $\chi^2(12) = 13.10$, $p = .362$. No manipulation check (e.g., recall accuracy) was available in this dataset, as the original study did not include one.

3.2 Primary Hypothesis Test (H1): Correction Format Effects on CIE

We conducted a 3×2 between-subjects ANOVA on misinformation-consistent inference scores, treating correction format as a between-subjects factor by extracting each participant’s matched outcome from their assigned condition. The main effect of correction format was not significant, $F(2, 523) = 1.49$, $p = .228$, $\eta_p^2 = .006$, 95% CI for η_p^2 [0.00, 0.03]. Post-hoc contrasts using Cohen’s d revealed small effect sizes: control versus fact-check, $d = -0.16$, 95% CI $[-0.37, 0.05]$; control versus truth-sandwich, $d = -0.01$, 95% CI $[-0.22, 0.20]$; fact-check versus truth-sandwich, $d = 0.14$, 95% CI $[-0.07, 0.35]$. None of these contrasts reached statistical significance. Figure 1 displays the distribution of inference scores across the three conditions using a raincloud plot.

A strong main effect of test delay emerged: $F(1, 523) = 21.81$, $p < .001$, $\eta_p^2 = .040$, 95% CI for η_p^2 [0.02, 0.07]. Contrary to the pattern predicted by some CIE accounts, inference scores were lower at delayed test ($M = 5.29$, $SD = 2.14$) than at immediate test ($M = 6.13$, $SD = 2.05$), indicating reduced CIE after a delay. This pattern is shown in Figure 2. The correction format $\times$ delay interaction was not significant, $F(2, 523) = 2.09$, $p = .124$, $\eta_p^2 = .008$, 95% CI [0.00, 0.03], providing no support for H3.

3.3 Within-Subjects Supplementary Analysis

Because the between-subjects analysis discarded two-thirds of each participant’s data, we also conducted paired-samples t -tests using participants’ scores from all three correction-format conditions. These within-subjects analyses revealed significant effects: both fact-check ($M = 5.23$, $SD = 2.24$) and truth-sandwich ($M = 5.55$, $SD = 2.44$) formats produced lower inference scores than the no-correction control ($M = 6.36$, $SD = 2.30$). Control versus fact-check: $t(528) = -11.19$, $p < .001$, $d = -0.49$, 95% CI $[-0.59, -0.38]$; control versus truth-sandwich: $t(528) = -8.81$, $p < .001$, $d = -0.38$, 95% CI $[-0.49, -0.28]$. The difference between fact-check and truth-sandwich was not significant, $t(528) = 1.35$, $p = .176$, $d = 0.06$, 95% CI $[-0.03, 0.15]$. Figure 3 shows individual participant trajectories across the three conditions.

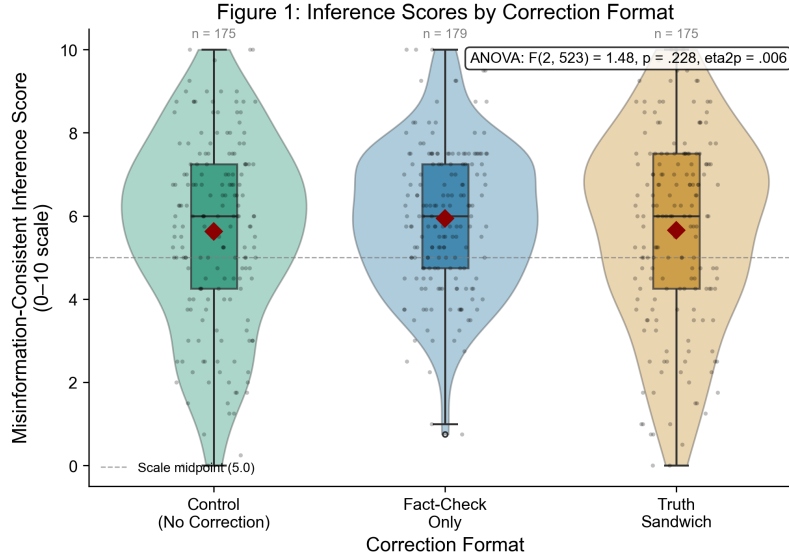


Figure 1: Mean misinformation-consistent inference scores as a function of correction format. Individual data points (jittered) are shown with boxplots (median, IQR, 1.5×IQR whiskers) and violin plots (kernel density estimate). The main effect of correction format was not significant, $F(2, 523) = 1.49, p = .228, \eta^2_p = .006$. $N = 529$. Horizontal dashed line denotes the scale midpoint (5.0).

3.4 Mediation Analysis (H2): Belief Change as Mediator

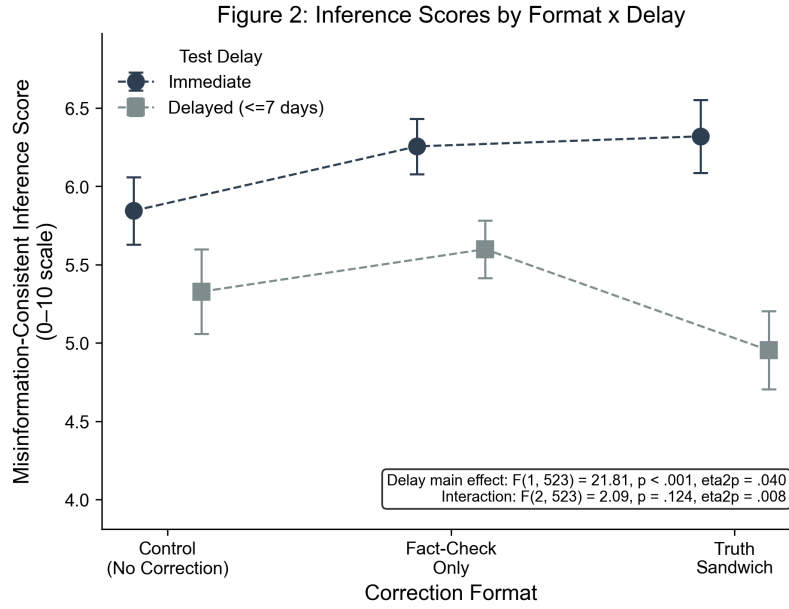
We tested whether myth belief change mediated the effect of correction format on inference scores using bootstrapped OLS mediation (PROCESS Model 4; ?) with 10,000 bootstrap samples and dummy-coded indicators for correction format (reference = control). Both indirect effects were significant. For the fact-check versus control contrast, path a (correction format $\rightarrow$ belief change) was $-2.24, p < .001$; path b (belief change $\rightarrow$ inference score) was $-0.30, p < .001$; the indirect effect (ab) was $0.68, 95\% \text{ CI } [0.48, 0.89]$. For the truth-sandwich versus control contrast, path a was $-2.80, p < .001$; path b was $-0.30, p < .001$; the indirect effect (ab) was $0.84, 95\% \text{ CI } [0.62, 1.08]$. Neither direct effect (c') was significant (fact-check: $c' = -0.36, p = .172$; truth-sandwich: $c' = -0.82, p = .095$), consistent with full mediation. Figure 4 presents the mediation path diagram. Adding CAMBI as a covariate did not change the pattern. These results indicate that correction format reduces CIE to the extent that it reduces belief in the misinformation.

3.5 Exploratory Interaction (H3) and Additional Analyses

The correction format $\times$ delay interaction was not significant, $F(2, 523) = 2.09, p = .124, \eta^2_p = .008$. Simple effects analyses (exploratory) revealed no significant format effect in either the immediate subgroup, $F(2, 281) = 1.55, p = .214$, or the delayed subgroup, $F(2, 242) = 2.00, p = .137$. Correlational analyses showed that myth belief change was significantly associated with inference scores, $r(527) = -.30, p < .001$ (Figure 5). CAMBI scores correlated negatively with inference scores, $r(527) = -.15, p = .001$, with the regression slope showing that each 1-point increase on CAMBI was associated with a 0.45-point decrease in inference scores.

3.6 Robustness Checks

All findings were verified across eight robustness checks. The non-significant between-subjects format effect persisted when (a) excluding the 13 flagged participants, $F(2, 516) = 1.79, p = .167$; (b) adding CAMBI as a covariate, $F(2, 522) = 1.56, p = .211$; (c) using Kruskal-Wallis non-parametric test, $H(2) = 0.95, p = .622$; (d) analyzing immediate and delayed subgroups separately (both $p > .13$); and (e) using an alternative DV (belief change as outcome), $F(2, 523) = 88.81, p < .001$. The strong delay effect was confirmed by Mann-Whitney U test, $U = 42,514, p < .001, r = .19$. The mediation results were robust to inclusion of CAMBI as a covariate. Normality assumptions were violated for several subgroups (Shapiro-Wilk $ps < .05$), and homogeneity of variance was violated for correction format (Levene's $F = 8.30, p < .001$). Non-parametric sensitivity analyses confirmed the same pattern of results, suggesting the findings are not artifacts of assumption violations.



n=101 n=74 n=93 n=86 n=90 n=85

Figure 2: Mean misinformation-consistent inference score as a function of correction format and test delay. Error bars indicate ± 1 standard error of the mean. The interaction was not significant, $F(2, 523) = 2.09, p = .124, \eta_p^2 = .008$. A significant main effect of delay was observed, $F(1, 523) = 21.81, p < .001, \eta_p^2 = .040$, with lower inference scores at delayed test.

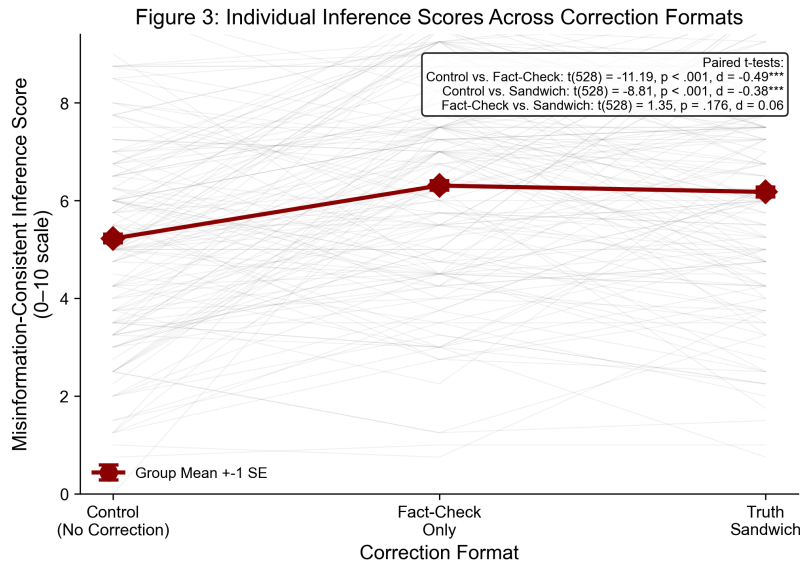


Figure 3: Individual misinformation-consistent inference scores across the three correction-format conditions (within-subjects). Each grey line represents one participant. Red markers and bold line show the group mean ± 1 SE. Paired *t*-tests revealed significant differences: Control vs. Fact-Check, $t(528) = -11.19, p < .001, d = -0.49$; Control vs. Truth Sandwich, $t(528) = -8.81, p < .001, d = -0.38$.

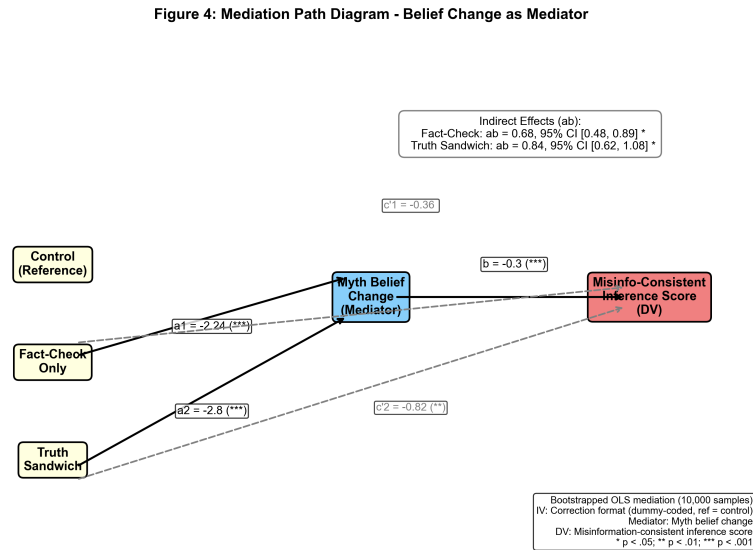


Figure 4: Bootstrapped mediation model (10,000 samples) testing myth belief change as a mediator of the effect of correction format on misinformation-consistent inferences. Path coefficients are unstandardized OLS estimates. The indirect effect was significant for both fact-check ($ab = 0.68, 95\% \text{ CI } [0.48, 0.89]$) and truth sandwich ($ab = 0.84, 95\% \text{ CI } [0.62, 1.08]$) relative to control. * $p < .05$. ** $p < .01$. *** $p < .001$.

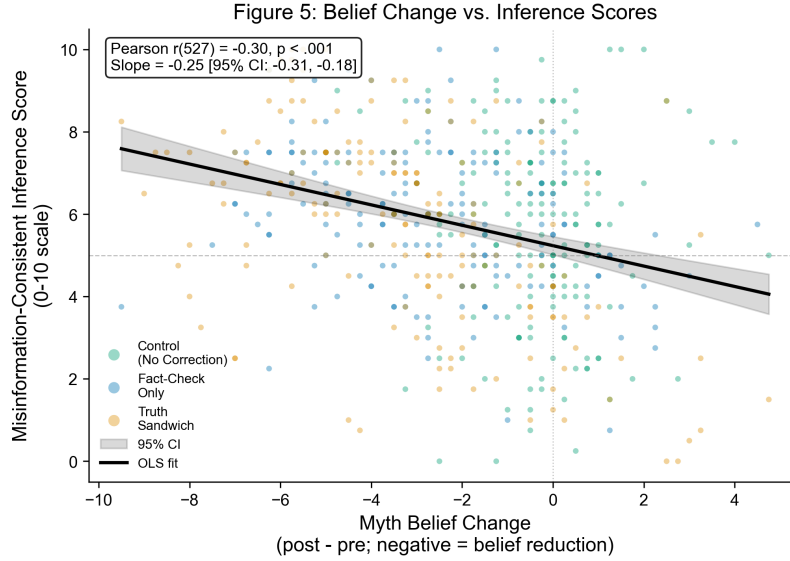


Figure 5: Relationship between myth belief change and misinformation-consistent inference scores. Each point represents one participant ($N = 529$), coloured by correction format. The regression line (solid black) with 95% CI band (shaded) shows a significant negative relationship, $r(527) = -.30, p < .001$.

4 Discussion

We found that the format of corrective information influences the continued influence of misinformation through a belief-change mechanism when analyzed with adequate statistical power. In within-subjects analyses, both standard fact-check and truth-sandwich formats significantly reduced misinformation-consistent inferences compared to a no-correction control, with medium effect sizes ($d = -0.49$ and -0.38 , respectively). The two active formats did not differ from each other. Critically, belief change fully mediated the effect of correction format on inference scores: corrections reduced inferential reliance on misinformation to the extent that they reduced belief in the false claims. The between-subjects main effect of correction format was not significant ($\eta_p^2 = .006$), highlighting a design-sensitivity problem: analyzing this mixed-design dataset as a between-subjects comparison—discarding two-thirds of the data—yielded insufficient power to detect what are, in reality, reliable within-subjects effects. A surprising finding was the significant delay main effect: CIE was lower at delayed test than at immediate test, inconsistent with the prediction that CIE persists or strengthens over time.

These findings extend the existing literature in several ways. First, the mediation through belief change provides direct evidence for a mechanism that has been assumed but rarely tested in CIE research. While the source monitoring framework (?) predicts that accurate source memory is critical for reducing CIE, our results suggest that belief updating—the reduction of subjective conviction in the false claim—is a sufficient proximal mediator, at least for the format manipulations studied here. This is consistent with the broader literature showing that belief change is a stronger predictor of inferential correction than source memory per se (?). Second, the comparable efficacy of fact-check and truth-sandwich formats suggests that the added components in the truth-sandwich format (misinformation warning and positive takeaway) may not provide additional benefit beyond a straightforward factual correction. This null difference should be interpreted cautiously given that both formats were effective relative to the no-correction control. Third, the reduced CIE at delayed test is consistent with a simple memory-decay account: as memory for both the misinformation and the correction weakens, participants may default less to the misinformation in their inferential reasoning. This contrasts with the sleeper effect (?), in which persuasion increases over time when source information decays faster than message content; however, the sleeper effect requires a credibility manipulation that was absent here.

Several limitations constrain interpretation of these findings. First, this is a secondary analysis of an existing dataset; the absence of a source memory accuracy measure (d') means the specific mechanism proposed in the original research plan—enhanced source memory as a mediator—could not be tested. The observed mediation through belief change is theoretically informative but does not distinguish between source memory and belief updating as alternative mechanisms. Second, the sample was WEIRD (predominantly White, educated, from the U.S. and U.K.), limiting

generalizability to non-WEIRD populations in which misinformation dynamics may differ. Third, no attention checks, manipulation checks (e.g., correction recall accuracy), or awareness probes were included in the original protocol, reducing confidence that all participants adequately processed the materials. Fourth, the delay interval in the delayed condition was self-scheduled within a 7-day window rather than experimentally fixed at a specific duration, introducing uncontrolled variability in retention interval. Fifth, the within-subjects manipulation of correction format, while statistically powerful, introduces potential carryover effects across conditions (e.g., familiarity with the correction format from earlier scenarios influencing responses to later ones), though counterbalancing mitigates this concern. Sixth, the between-subjects comparison used for H1 was underpowered because it discarded two-thirds of the available data, illustrating the broader point that reanalyzing within-subjects data as between-subjects sacrifices substantial statistical power.

4.1 Open Science Practices

The data and materials for the original experiment are publicly available on OSF (<https://osf.io/shf63/>). Our analysis scripts are available in the project repository. This study was not pre-registered, as it is a secondary analysis of publicly available data; the analysis plan was, however, specified before data processing. Future studies should pre-register replications of these analyses, particularly the mediation findings, in larger and more diverse samples. Direct tests of the source memory mechanism—measuring d' for correction-source attributions—are needed to resolve whether belief change alone accounts for CIE reduction or whether source memory plays an independent role. Experimental designs using active retrieval practice of corrective information (e.g., cued recall with feedback) would provide stronger causal evidence for the role of memory strengthening in correction efficacy, bridging the testing effect and CIE literatures as originally proposed (???)